COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE2124
DATE PERFORMED:	07/29/26	**DATE SUBMITTED:**	07/29/26
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ACTIVITY NO. ___ [ACTIVITY NAME]

1. PROCEDURE OUTPUT

Progress:ILOA

```cpp
#include <iostream>
#include <stack>

int main() {

    std::stack<char> stack1;

    std::cout << "Is the stack empty? " << stack1.empty() << std::endl;

    stack1.push('k');
    std::cout << "Current top of the stack: " << stack1.top() << std::endl;

    stack1.push('e');
    std::cout << "Current top of the stack: " << stack1.top() << std::endl;

    stack1.push('n');
    std::cout << "Current top of the stack: " << stack1.top() << std::endl;

    stack1.push('n');
    std::cout << "Current top of the stack: " << stack1.top() << std::endl;

    stack1.push('e');
    std::cout << "Current top of the stack: " << stack1.top() << std::endl;

    stack1.push('t');
    std::cout << "Current top of the stack: " << stack1.top() << std::endl;

    stack1.push('h');
    std::cout << "Current top of the stack: " << stack1.top() << std::endl;

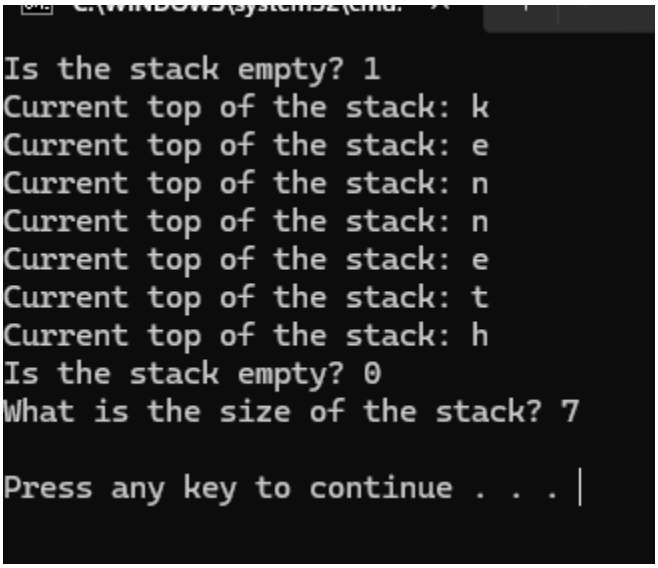
    std::cout << "Is the stack empty? " << stack1.empty() << std::endl;
    std::cout << "What is the size of the stack? " << stack1.size() << std::endl;

    //swap

    //emplace


    return 0;
}
```

Output:


Is the stack empty? 1
Current top of the stack: k
Current top of the stack: e
Current top of the stack: n
Current top of the stack: n
Current top of the stack: e
Current top of the stack: t
Current top of the stack: h
Is the stack empty? 0
What is the size of the stack? 7
Press any key to continue . . .

Progress:ILOAB1

ILOAB1 > main.cpp > push()

```cpp
#include <iostream>

#define maxSize 10

int stackArr[maxSize];
int top = -1, newData, choice;

bool isEmpty();
bool isFull();
void Top();
void push();
void pop();

int main() {

    std::cout << "================================\n";
    std::cout << "Stack Operations:\n";
    std::cout << "[1] Push [2] Pop [3] isEmpty [4] isFull [5] Top\n";
    std::cout << "Enter your choice: ";
    std::cin >> choice;   // Added semicolon

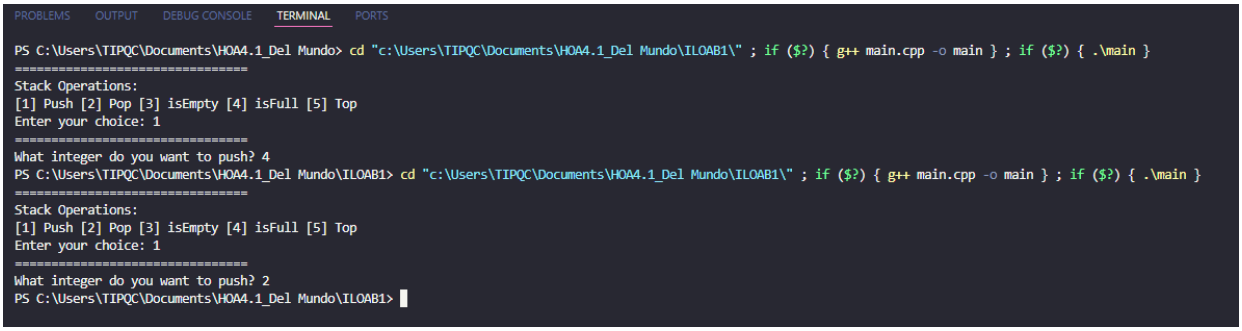
    std::cout << "================================\n";

    switch (choice) {
        case 1:
            push();
            break;
        case 2:
            pop();
            break;
        case 3:
            std::cout << "Is the stack empty? " << isEmpty() << std::endl;
            break;
        case 4:
            std::cout << "Is the stack full? " << isFull() << std::endl; // Fixed stdd
            break;
        case 5:
            Top(); // Fixed ; to :
            break;
        default:
            std::cout << "Invalid Choice!" << std::endl;
            break;
    }

    return 0;
}
```

```
47
48  bool isEmpty() {
49      return (top == -1);
50  }
51
52  bool isFull() {
53      return (top == maxSize - 1);
54  }
55
56  void Top() {
57      if (isEmpty()) {
58          std::cout << "Stack is empty!" << std::endl;
59          return;
60      }
61
62      std::cout << "The top element is: " << stackArr[top] << std::endl;
63  }
64
65  void push() {
66
67      if (isFull()) {
68          std::cout << "Stack is full!" << std::endl;
69          return;
70      }
71
72      std::cout << "What integer do you want to push? ";
73      std::cin >> newData;
74
75      stackArr[++top] = newData;
76  }
77
78  void pop() {
79
80      if (isEmpty()) {
81          std::cout << "Stack is empty!" << std::endl;
82          return;
83      }
84
85      std::cout << "Popping: " << stackArr[top] << std::endl;
86
87      top--;   // Added semicolon
88  }
```

Output:


PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
PS C:\Users\TIPQC\Documents\HOA4.1_Del Mundo> cd "c:\Users\TIPQC\Documents\HOA4.1_Del Mundo\ILOAB1\" ; if ($?) { g++ main.cpp -o main } ; if ($?) { .\main }
==============================
Stack Operations:
[1] Push [2] Pop [3] isEmpty [4] isFull [5] Top
Enter your choice: 1
==============================
What integer do you want to push? 4
PS C:\Users\TIPQC\Documents\HOA4.1_Del Mundo\ILOAB1> cd "c:\Users\TIPQC\Documents\HOA4.1_Del Mundo\ILOAB1\" ; if ($?) { g++ main.cpp -o main } ; if ($?) { .\main }
==============================
Stack Operations:
[1] Push [2] Pop [3] isEmpty [4] isFull [5] Top
Enter your choice: 1
==============================
What integer do you want to push? 2
PS C:\Users\TIPQC\Documents\HOA4.1_Del Mundo\ILOAB1>

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILOA > main.cpp > ...

```cpp

#include <iostream>
#include <stack>


int main (){
    std::stack<int> stack1;
    std::stack<int> stack2;
    std::cout<<"Testing the stack STL"<<std::endl;


    std::cout<<"Is the Stack Empty? "<<stack1.empty()<<std::endl;

    stack1.push(10);
    std::cout<<"The top of the stack is "<<stack1.top()<<std::endl;
    stack1.push(9);
    std::cout<<"The top of the stack is "<<stack1.top()<<std::endl;
    stack1.push(8);
    std::cout<<"The top of the stack is "<<stack1.top()<<std::endl;
    stack1.push(7);
    std::cout<<"The top of the stack is "<<stack1.top()<<std::endl;
    stack1.pop();
    std::cout<<"The top of the stack will be popped out"<<std::endl;
    std::cout<<"The top of the stack is "<<stack1.top()<<std::endl;
    std::cout<<"The Size of the Stack is "<<stack1.size()<<std::endl;
    stack1.emplace(6);
    std::cout<<"The top will be replaced by "<<stack1.top()<<std::endl;

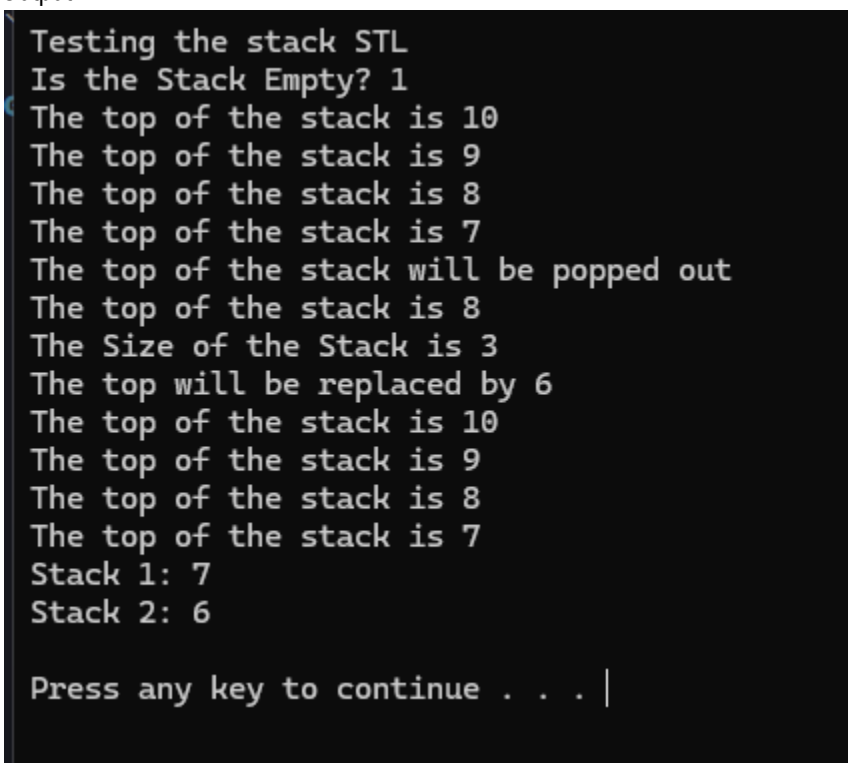

    stack2.push(10);
    std::cout<<"The top of the stack is "<<stack2.top()<<std::endl;
    stack2.push(9);
    std::cout<<"The top of the stack is "<<stack2.top()<<std::endl;
    stack2.push(8);
    std::cout<<"The top of the stack is "<<stack2.top()<<std::endl;
    stack2.push(7);
    std::cout<<"The top of the stack is "<<stack2.top()<<std::endl;


    stack1.swap(stack2);
    std::cout<<"Stack 1: "<<stack1.top()<<std::endl;
    std::cout<<"Stack 2: "<<stack2.top()<<std::endl;


    return 0;
}

```

Output:


Testing the stack STL
Is the Stack Empty? 1
The top of the stack is 10
The top of the stack is 9
The top of the stack is 8
The top of the stack is 7
The top of the stack will be popped out
The top of the stack is 8
The Size of the Stack is 3
The top will be replaced by 6
The top of the stack is 10
The top of the stack is 9
The top of the stack is 8
The top of the stack is 7
Stack 1: 7
Stack 2: 6
Press any key to continue . . .

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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